Verification-First Legal RAG: A Five-Rail Closing Audit and Calibration Stress Test on Swiss Federal Law

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Abstract

We propose a five-rail closing audit pipeline for retrieval-augmented legal LLMs — case-citation existence, statute-reference resolution, verbatim-quote source-matching, decision-date sanity, and proposition grounding by a separate-call LLM judge — and stress-test it against Claude Sonnet 4.6 on 30 Swiss-federal-law questions in a *prior-only* condition (no retrieval). Four rails are deterministic (∼3 ms per draft); the fifth is judge-mediated (∼2 s, ≈\$0.005 per call). We report two runs against the same questions and benchmark artefact: a v1 run, and a v2 run after deploying mechanical fixes (citation-leading claim extraction, judge-call retry) that the v1 run identified as audit-pipeline gaps.

Three findings. (i) *Citation existence is not the bottleneck*: in v1 the model emits 33 case citations and fabricates 1 (3.0 %); in v2 it emits 37 and fabricates 1 (2.7 %); in both runs the same docket is fabricated (BGE/DTF 117 II 198 on q-017). In v1, 5 of the 6 wrong drafts contain only valid citations. Hallucination rates reported by Dahl et al. 2024 (58–82 %) and Magesh et al. 2025 (17–33 %) were measured under different model and condition combinations and we do not claim a like-for-like comparison. (ii) *Calibration of the grounding judge is the open problem*: with the mechanical fixes in place, the v2 grounding rail evaluates 13 of 30 drafts (vs. 8/30 in v1) and fires on 3 of the 6 wrong drafts (50 % wrong-draft flag rate for the rail alone) but also on 10 of the 24 correct drafts (FPR 41.7 %). Inspection shows the three wrong-draft flags are clean semantic catches on the misrepresented-holding class, but the judge is opinionated enough on correct drafts that running it as a binary gate would over-flag. The full v2 audit (case + statute + quote + date + grounding) flags 4 of 6 wrong drafts (the quote rail catches q-025's non-verbatim French quotation) at FPR 50 %. (iii) *End-to-end run variance is substantial*: at TEMPERATURE = 0, the generator produces a different draft for all 30 questions across our two runs, and 4 of the 30 receive different c_{eli} verdicts (wrong-set {q-017, q-018, q-021, q-025, q-026, q-030} in v1, {q-006, q-018, q-021, q-025, q-028, q-030} in v2). We do not isolate generator variance from judge variance in the present paper because both compound under the API conditions that produce the published numbers; v1.0 will report inter-run agreement separately for held-fixed-draft judging and end-to-end re-runs, and substitute an independent judge. We release the audit pipeline (MIT) and **Swiss Legal RAG Bench v0.2** (30 author-curated questions, DE / FR / IT; benchmark packaging CC0).

1 Introduction

Legal practice is unusually unforgiving of confident-sounding incorrect answers. A lawyer who cites a fabricated case, miscites a statute, or attaches a false holding to a real authority risks professional discipline and a wrong outcome for the client; in Switzerland, this is a violation of the *Sorgfaltspflicht* (duty of care) under the federal *Anwaltsgesetz* (BGFA, SR 935.61, art. 12). As large language models become a routine part of legal research, this asymmetry collides with a measured failure mode: 58–82 % of legal queries to general-purpose LLMs (Dahl et al., 2024) and 17–33 % of queries to commercial legal-RAG products (Magesh et al., 2025) produce a fabricated authority, miscited statute, or unsupported quotation.

Two recent developments shape the present work. First, Butler & Butler (2026) introduce a clean methodology for evaluating legal-RAG systems: decompose each error into three orthogonal binary signals (correctness c_{eli}, groundedness g_{eli}, retrieval accuracy r_{eli}) and read off the failure class from their joint distribution. Their methodology lets us ask *which* rail of a verification system addresses *which* error class. Second, the Swiss legal NLP literature has matured rapidly: SCALE (Rasiah et al., 2023) introduced a multi-task Swiss legal benchmark suite, the Niklaus group released anonymization tooling (Niklaus, Mamié et al., 2023) and re-identification risk measurement (Nyffenegger, Stürmer & Niklaus, 2024) for Swiss court rulings, SwiLTra-Bench (Niklaus et al., 2025) provided 180 k aligned Swiss legal translation pairs, and the SLDS dataset (Rolshoven, Rasiah et al., 2025) offered 20 k cross-lingual SFSC summarisation pairs. None of these target retrieval-augmented *generation* as the unit of evaluation; the present paper occupies that specific gap.

Two findings, in tension, motivate the rail decomposition. Our central empirical result is that on a closed Swiss-law corpus (969 k decisions, daily refresh; portions of which are plausibly present in the web-scale pretraining of the model under test) Claude Sonnet 4.6 fabricates only **3.0 %** of emitted citations. Hallucination rates reported by Magesh et al. (2025, 17–33 % on commercial legal-RAG products) and Dahl et al. (2024, 58–82 % on general LLMs) were measured under different model and condition combinations and we do not claim a like-for-like comparison. Citation existence is not the bottleneck in our setting. The c_{eli} correctness rate is 80 % (24/30) in both of our runs, but the wrong-draft *set* differs by 4 of 6 across runs — an end-to-end-run noise floor (combining generator variance and judge variance, both at TEMPERATURE = 0) we surface as a primary limitation in §6. The dominant residual error class is *real citation, misrepresented holding*: the model cites a real BGE correctly but omits or misrepresents its load-bearing precedential element. Per-citation existence checks cannot detect this class; an LLM-judge grounding rail can, but the question of how to *calibrate* that judge — when to defer to the draft, when to override — becomes the dominant open problem once the rail's mechanical claim-extraction gap is closed.

The conceptual move. A production verification system for legal RAG should not be a single black-box LLM judge over the final draft, nor a single per-citation retrieval check, but a **decomposition into independent rails, one per named error class**, each implemented with the cheapest mechanism that can detect that class. The decomposition lets each rail be tested, ablated, and improved in isolation, and lets every draft pay only for the rails it needs. We instantiate the design as a five-rail closing audit pipeline (§3) deployed in production at `mcp.opencaselaw.ch`.

Contributions.

1. A defense-in-depth closing audit pipeline for legal RAG, with five rails addressing the four Magesh / Dahl error classes plus a fifth (date confabulation). Latency ≤ 3 ms for the four deterministic rails; ~ 2 s, $\approx$\$0.005 per call for the LLM-judge rail.

2. **Swiss Legal RAG Bench v0.2**: a 30-question *pilot multilingual evaluation set* for Swiss federal law (DE / FR / IT; single-curator), modelled on Butler & Butler (2026) with one cross-lingual retrieval question (q-028) as a placeholder for a future cross-language suite.

v1.0 (150 questions, named expert co-authors, multi-annotator protocol) is the planned gold-standard deliverable.

3. Per-rail ablation across two runs — v1 against the as-instrumented audit code, v2 after deploying the mechanical fixes (citation-leading claim extraction, judge-call retry) the v1 inspection identified — identifying *calibration* of the grounding judge as the open problem. In v2 the grounding rail evaluates 13 of 30 drafts (vs. 8 in v1) and makes 3 clean semantic catches on wrong drafts (q-006, q-018, q-021) but also fires on 10/24 correct drafts (FPR 41.7 %). Two wrong drafts (q-028, q-030) reach the judge but the extracted claim is a narrow factual statement that is literally supported by the source even though the draft as a whole is wrong — the *narrow-claim, broad-error* failure mode joins calibration as the second open problem.

2 Related Work

Legal-LLM hallucination measurement. Dahl et al. (2024, *Large Legal Fictions*) profiled hallucination on six legal-task types in general-purpose LLMs, reporting fabricated-authority rates of 58–82 %. The follow-up Magesh et al. (2025, *Hallucination-Free?*) measured commercial legal-RAG tools (Lexis+ AI, Westlaw AI-Assisted Research, Ask Practical Law) at 17–33 %. Our paper accepts these as the problem statement; the five-rail pipeline of §3 is the treatment.

Legal-RAG benchmarks. Butler & Butler (2026, *Legal RAG Bench*) introduced the three-signal error decomposition (c_{eli}, g_{eli}, r_{eli}) we extend. Their benchmark covers Australian criminal law (4,876 passages from the Victorian Criminal Charge Book, 100 hand-crafted questions) and uses a closed corpus. We adapt the methodology to Swiss multilingual federal law, adding a cross-language retrieval signal because Switzerland is unusual among major jurisdictions in that the authoritative version of a leading case may be in a language other than the question's (we acknowledge that v0.2 includes only one cross-lingual question, q-028; v1.0 expands this to a full cross-language retrieval suite). Closed-domain multiple-choice benchmarks (LegalBench-RAG, Pipitone & Houir Alami 2024; HousingQA / BarExamQA, Zheng et al. 2025; LegalBench, Guha et al. 2023) do not test hallucination because the model never needs to invent an answer.

Swiss legal NLP. Recent Swiss legal NLP has matured rapidly around the Niklaus / Stürmer / Stanford / UZH / Bern axis. Three lines are directly adjacent to ours. (i) *Privacy and anonymisation*: Niklaus, Mamié et al. (2023) released BERT-based models for entity-level anonymisation of Swiss FSCS rulings, improving the operational *Anom2* system used by the Federal Supreme Court; Nyffenegger, Stürmer & Niklaus (2024) systematically measured LLM-based re-identification risk on anonymised Swiss court rulings, finding the risk minimal for top models on *court decisions* (higher on Wikipedia entities). We discuss the implications for our corpus exposure in §6. (ii) *Multilingual benchmarks*: SwiLTra-Bench (Niklaus et al., 2025) is a 180 k-pair Swiss legal translation benchmark across DE / FR / IT / RM / EN, introducing the SwiLTra-Judge specialised LLM evaluator with human-expert validation. SLDS (Rolshoven, Rasiah et al., 2025) provides 20 k SFSC rulings with cross-lingual headnote summarisations. We do not duplicate these tasks; our cross-lingual test (q-028) targets retrieval, not translation or summarisation. (iii) *Multi-task suites*: SCALE (Rasiah et al., 2023) covers citation extraction, court-view generation and summarisation; Swiss-Judgment-Prediction (Niklaus, Chalkidis & Stürmer, 2021) addresses outcome prediction; MultiLegalPile (Niklaus et al., 2023) provides a 689 GB multilingual legal pre-training corpus. None of these target retrieval-augmented *generation evaluation*; the present paper occupies that specific gap.

Verification methods. Self-RAG (Asai et al., 2023) and CRAG (Yan et al., 2024) propose general-domain self-verification or retrieval-correction; both treat verification as *pre-generation* (judge passage quality before generating). Our pipeline is *post-generation* verification: parse the generated draft, audit each emitted reference against the closed corpus, return a verdict per rail. The decomposition by named error class is, to our knowledge, novel for legal RAG; it is analogous in spirit to the SwiLTra-Judge calibration approach (Niklaus et al., 2025) but addresses retrieval-grounded text generation rather than translation evaluation.

Positioning. To our knowledge, no prior work proposes a *deployed* mechanism that systematically reduces hallucination by decomposing the verification problem into independent error-class detectors. Existing RAG systems we are aware of treat verification as either a single LLM-judge pass over the final output (which inherits the same model's biases) or per-citation retrieval validation (which catches existence errors but not propositional mis-grounding). The five-rail closing audit we propose in §3 is, to our knowledge, the first to address each error class with the cheapest mechanism that can detect it, with each rail independently testable and ablatable.

3 The Five-Rail Audit Pipeline

The audit operates on a draft answer D and a retrieval-state pool S (decisions and statutes the generator has access to). It returns a list of issues $\mathcal{I}(D, S)$, each labelled with one of five categories (Figure 1). An OK verdict (no issues) means the audit found nothing; ¬OK means at least one rail fired. The five rails operate *in parallel* on the same draft.

LLM draft D $\Longrightarrow$

Rail 1: case-citation existence (~ 2 ms, deterministic)
Rail 2: statute-reference resolution (~ 1 ms, deterministic)
Rail 3: verbatim-quote source-matching (~ 2 ms, deterministic)
Rail 4: decision-date sanity (< 1 ms, deterministic)
Rail 5: proposition grounding (opt-in) (~ 2 s, judge)

$\Longrightarrow$ **verdict** $\mathcal{I}(D, S)$

Figure 1: Five-rail closing audit pipeline. Four deterministic rails address citation-, statute-, quotation-, and date-level fabrications; an opt-in LLM-judge rail addresses proposition-grounding errors. Each rail is independently testable and ablatable; the configuration adopted by a deployment is a per-rail subset.

Rail 1 — case-citation existence. A regex parses the draft for Swiss case-citation forms (BGE / ATF / DTF, BGer / TF dockets, BVGer / BStGer / BPatGer dockets, MKGE numerical references). Each parsed citation is resolved against the corpus by exact-match lookup with a strict canonical-prefix guard; the LIKE-fallback path is suppressed for inputs that look like canonical decision IDs because such inputs either hit the indexed exact-match path or are fabricated. Latency: 2–5 ms per draft regardless of citation count. Catches: invented BGE numbers, mis-typed docket numbers, citations to retracted decisions. Optional pinpoint check: if the citation includes E. $n.m$ or consid. $n.m$, the pinpoint must match either the structured-extraction sidecar or the body-text `regeste` of the cited decision.

Rail 2 — statute-reference resolution. A second regex parses for Art. N LAW forms (with German / French / Italian abbreviation synonyms) and resolves each against the Fedlex SPARQL mirror cached locally as `statutes.db`. The resolution is two-step — abbreviation → SR number → article text — and is cached at process scope. Latency: 1–3 ms per draft. Catches: fabricated law abbreviations, wrong article numbers, references to repealed articles.

Connector words (“und”, “oder”, “et”, “ou”) are case-sensitively filtered to prevent the regex eating them as fake law codes after Abs. N structures.

Rail 3 — verbatim-quote source-matching. Quoted substrings of ≥ 30 characters in the draft are extracted (German „...“, French « ... », Italian « ... », English curly “ . . . ”, ASCII “...”) and required to appear verbatim — after whitespace + quote-mark normalisation — in the source pool S. The pool comprises the `regeste` text plus first 8 k characters of each cited decision, plus the article text of each cited statute. Latency: ∼2 ms. Catches: hallucinated quotations, paraphrases inside quotation marks.

Rail 4 — decision-date sanity. For each verified case citation, parse adjacent vom / *du* / *del* DD.MM.YYYY constructs within 60 characters of the citation span and compare to the stored `decision_date`. Catches: model emits a real BGE but pairs it with a wrong date, suggesting confusion between two cases with similar dockets. Latency: < 1 ms.

Rail 5 — proposition grounding (opt-in). For each verified citation, a claim sentence is extracted: first the immediately preceding sentence (up to 250 characters before the citation, terminated by sentence boundary), and if that yields no ≥20-character candidate, the rest of the citation’s own sentence (forward fallback up to 500 characters, used when the citation leads its sentence in patterns like “In BGE X, ...” / “Selon ATF Y, ...”). “See”-cite markers “vgl.” / “siehe” / “cf.” / “cfr.” suppress the rail; the judge call itself is wrapped in a 3-attempt retry with exponential backoff (1 s, 2 s) so transient API failures do not silently surface as `judge_unavailable`. All (claim, source) pairs go in a *single batched* prompt to a separate Sonnet judge call (same model family as the generator; see §6) that returns a JSON array of verdicts {`yes, partial, no, contradicts, unrelated`}. Verdicts in {`no, contradicts, unrelated`} become issues. One Sonnet call per draft regardless of citation count. Latency: 1.5–2.5 s. Cost: ≈ $0.005 per call. Targets the Butler / Butler “reasoning error” class (real citation + misrepresented holding); empirical catch rate on this class is qualified in §5 (the rail in our experiment fires on secondary attachment issues rather than cleanly on the underlying mischaracterisation).

Output canonicalisation. When all rails pass, the audit emits a `linked_text` field with every parsed citation replaced by its corpus-canonical form wrapped in a Markdown link to the source URL on `mcp.opencaselaw.ch/entscheid/<id>`. Drafts arrive with messy citation spacing or case and leave publication-ready.

4 Swiss Legal RAG Bench v0.2

We release Swiss Legal RAG Bench v0.2 alongside this paper as a *pilot multilingual evaluation set*. Version 0.2 contains 30 questions *author-curated by the maintainer of* `opencaselaw.ch` *(the corpus mirror underlying the audit pipeline)*, drawn from the major substantive areas of Swiss federal law (Schuldrecht, Sachenrecht, Familienrecht, Erbrecht, Strafrecht, Strafprozess, Verfassungsrecht, Verwaltungsrecht, Mietrecht, Arbeitsrecht, Sozialversicherungsrecht, Verkehrsrecht, Aktienrecht, SchKG, Datenschutz, Verjährung, Wirtschaftsfreiheit; 17 distinct areas). The language distribution is DE = 18, FR = 8, IT = 4 (60 / 27 / 13 %). v0.2 is explicitly framed as a single-curator seed; v1.0 will adopt a multi-annotator protocol with named expert co-authors from Swiss law faculties (see §6).

Question construction. Each question is paired with a concise reference answer (1–4 sentences in the question’s language) and annotated supporting evidence (statute SR-number + article number + language; leading-case `decision_id` (s) where applicable). Of the 30 questions, 16 (53 %) are anchored on a verified leading-case BGE; 14 (47 %) carry the `doctrine_from_bge` claim type, in which the expected answer derives a doctrinal proposition from a seed BGE rather than restating its holding directly. The two annotations are not disjoint: every `doctrine_from_bge` question is also anchored on a leading-case BGE. The split exists to let the ablation surface where each rail contributes — case-existence on the seed citations, the grounding judge on the doctrinal proposition.

Corpus verification of evidence. Every `decision_id` cited as evidence was resolved against the live corpus (`mcp.opencaselaw.ch/api/decisions/<id>`) before the benchmark was published; every BGE returned a 200 response with non-zero citation count and an on-topic `regeste`. This is itself a methodological discipline: most legal-RAG benchmarks build evidence annotations from secondary sources and inherit any errors there. We commit to corpus-first construction, with the audit trail published in the supplementary material.

Cross-lingual extension. Switzerland is unusual among major civil-law jurisdictions in that the authoritative version of a leading case may be in a language other than the question's. Question q-028 (IT) anchors on DTF 126 III 113 (which has regesti in DE / FR / IT) to test the cross-language retrieval claim. v1.0 expands this dimension to a full cross-language retrieval suite.

Versioning. v0.2 is a seed sufficient to characterise the audit pipeline in this paper. **v1.0** is targeted at 150 questions with named expert co-authors from Swiss law faculties and a multi-annotator protocol (inter-annotator agreement reported, third-party adjudication for disagreements). We explicitly invite contributors for v1.0.

5 Experiments

Setup. We use Claude Sonnet 4.6 (released 2026-02-17) as both the generator and the judge for c_{eli}, g_{eli}, and the in-pipeline grounding rail. Using one model family on both sides is a known limitation (§6); v1.0 will report inter-judge agreement against an independent model. All 30 questions in v0.2 are run in a *prior-only* condition (no retrieval): the generator sees only the question text and must rely on training knowledge. We chose the prior-only condition as an audit-pipeline *stress test*, not as an end-to-end RAG evaluation: it maximises the density of recoverable errors, and the no-retrieval setup is methodologically aligned with Dahl et al. 2024's general-LLM hallucination measurement (we do not claim it replicates Dahl's substantive coverage of US legal tasks; our questions are Swiss federal law, single-curator). A retrieval-augmented condition runs in production at `mcp.opencaselaw.ch` (BM25 / RRF / Haiku rerank, see §6); end-to-end RAG evaluation against v1.0 is forthcoming work.

For each question we collect: the prior-only draft, the per-category issue counts from `attest_response` with all five rails on, the c_{eli} judge verdict against the v0.2 reference answer, and the g_{eli} judge verdict against the empty source pool. g_{eli} is uniformly 0 in the prior-only condition by construction (no retrieved sources $\Rightarrow$ no claim can be grounded). We use the c_{eli} verdict as ground truth for the per-rail ablation.

Per question we issue four base Sonnet calls (one generation, one in-pipeline grounding judge, one c_{eli} judge, one g_{eli} judge) plus one deterministic-rails pass; the in-pipeline grounding judge can additionally retry up to twice on transient API failure (we observed 1 such retry across the v2 run on q-013). The 6-row ablation in Table 1 is computed in post-processing by intersecting each draft's per-rail fire flags with each configuration's enabled rail set. Total: $\sim$120 Sonnet calls across 30 drafts; v1 wall-clock 106 s and v2 wall-clock 104 s with 4-way parallelism; <\$2 in API charges per run.

Citation-level fabrication rate. Across the 30 prior-only drafts, Claude Sonnet 4.6 emits 33 case citations in v1 (32 valid, fabrication **3.0 %**) and 37 in v2 (36 valid, fabrication **2.7 %**). In both runs the single fabricated docket is on q-017 (BGE/DTF 117 II 198, an invented combination of a real volume number and an unrelated case position). More informatively, in v1 5 of the 6 drafts judged wrong by the c_{eli} judge contain only valid citations: *citation existence is not the bottleneck in this setting.* Magesh et al. (2025) reported 17–33 % on commercial legal-RAG tools and Dahl et al. (2024) reported 58–82 % on general LLMs; those measurements were taken under different model families, query distributions, and tool conditions than ours, so the absolute numbers should not be read as a like-for-like comparison. Two non-exclusive hypotheses, neither tested here: (i) Swiss BGE references appear at scale on the open web — including via our own published Hugging Face dataset — and the model may have internalised the citation conventions; (ii) the prior-only condition asks short, well-bounded questions where the model can refuse, hedge, or ground in widely-known leading cases, whereas Magesh's tools were exercised on more open-ended litigation queries.

Per-rail ablation: two runs. Of the 30 prior-only drafts, 24 are judged correct ($c_{eli} = 1$) and 6 wrong ($c_{eli} = 0$) in each run by the second-pass Sonnet judge (same model family as the generator; see §6). We report two runs: **v1** against the original audit code at commit `4553379`, and **v2** after deploying the mechanical fixes (citation-leading claim extraction, judge-call retry) at commit `12d32d3`. Both runs use the same 30 questions, the same generator, the same judge model, and `temperature`=0. Table 1 reports the v2 numbers as primary; Table 2 reports the v1 baseline for comparison. We deliberately call the wrong-draft column **WFR** (wrong-draft flag rate) rather than "TPR" or "catch rate" because, as the inspection later in this section shows, several flags happen to land on wrong drafts without semantically detecting the underlying primary error.

Table 1: **v2 (post-mechanical-fix) ablation.** Per-rail-configuration flag rates, v0.2 prior-only condition, audit code at commit `12d32d3`. $n = 30$ drafts; $c_{eli} = 0$ for 6 (wrong: {q-006, q-018, q-021, q-025, q-028, q-030}) and $c_{eli} = 1$ for 24 (correct). WFR (wrong-draft flag rate) $= P(\text{any enabled rail fires} \mid c_{eli} = 0)$, computed on $n = 6$; FPR $= P(\text{any enabled rail fires} \mid c_{eli} = 1)$, computed on $n = 24$; Net = WFR − FPR. WFR counts *any* rail firing on a wrong draft and does *not* certify that the flag identifies the draft's primary error.

Config	Rails enabled	WFR	WFR 95% CI	FPR	Net
C_0	∅ (no rails)	0/6 = 0.0 %	[0.0, 39.0]	0.0 %	+0.0 pp
C_1	+ case	0/6 = 0.0 %	[0.0, 39.0]	4.2 %	−4.2 pp
C_2	+ statute	0/6 = 0.0 %	[0.0, 39.0]	12.5 %	−12.5 pp
C_3	+ quote	1/6 = 16.7 %	[3.0, 56.4]	12.5 %	+4.2 pp
C_4	+ date (4 deterministic)	1/6 = 16.7 %	[3.0, 56.4]	12.5 %	+4.2 pp
C_5	+ grounding (full 5-rail)	4/6 = 66.7 %	[30.0, 90.3]	50.0 %	+16.7 pp

Three v2 findings emerge.

First, in v2 the case-existence rail no longer holds the perfect-precision floor it held in v1. The v1 wrong set contained q-017 (a fabricated DTF 117 II 198) which C_1 caught cleanly. In v2, the c_{eli} judge re-scored q-017 as *correct* (the draft's substantive flaws — omission of cpv. 3, mischaracterisation of unjustified-refusal consequences — did not change between runs, but the judge's tolerance for them did; this is the run-to-run variance we discuss in §6). In v2 the deterministic rails fire only on correct-judged drafts (case rail: 1, statute: 2; FPR by C_2: 12.5 %, WFR 0 %). The *quote* rail does pick up q-025: the DE draft embedded the French phrase « coût indirect de la prise en charge » in quotation marks but the cited source contains no such verbatim string, so the rail correctly flags the quotation as unsupported. That single

Table 2: **v1 (pre-mechanical-fix) ablation, for comparison.** Audit code at commit `4553379`; same 30 questions, same generator, same judge model, TEMPERATURE $= 0$. c_{eli} wrong set in v1: {q-017, q-018, q-021, q-025, q-026, q-030} — 4 of these labels flipped between v1 and v2 (q-017 and q-026 went WRONG → CORRECT; q-006 and q-028 went CORRECT → WRONG), which we discuss as a primary limitation in §6. The C_5 jump in WFR (3/6 in v1 → 4/6 in v2) and the FPR jump (25 % in v1 → 50 % in v2) both reflect the mechanical fixes unblocking all 9 of v1's `skipped_no_claim` drafts and the 2 `judge_unavailable` drafts; one new judge error appeared in v2 on q-013.

Config	Rails enabled	WFR	WFR 95% CI	FPR	Net
C_0	$\emptyset$ (no rails)	$0/6 = 0.0\,\%$	$[0.0, 39.0]$	$0.0\,\%$	$+0.0$ pp
C_1	+ case	$1/6 = 16.7\,\%$	$[3.0, 56.4]$	$0.0\,\%$	$+16.7$ pp
C_2	+ statute	$1/6 = 16.7\,\%$	$[3.0, 56.4]$	$4.2\,\%$	$+12.5$ pp
C_3	+ quote	$1/6 = 16.7\,\%$	$[3.0, 56.4]$	$8.3\,\%$	$+8.3$ pp
C_4	+ date (4 deterministic)	$1/6 = 16.7\,\%$	$[3.0, 56.4]$	$8.3\,\%$	$+8.3$ pp
C_5	+ grounding (full 5-rail)	$3/6 = 50.0\,\%$	$[18.8, 81.2]$	$25.0\,\%$	$+25.0$ pp

hit raises WFR from 0 % to 16.7 % at C_3.

Second, the grounding rail in v2 is now actually evaluating drafts. In v1, 9 drafts were `skipped_no_claim` (q-020, q-021, q-022, q-024, q-025, q-026, q-027, q-028, q-029) because the claim-extraction heuristic looked only backwards from the citation and found nothing of substance — a structural bias against the citation-leading pattern ("In BGE X, ...", "Selon ATF Y, ...") that is very common in Swiss legal writing. The v2 audit code adds a forward fallback that extracts the claim from the rest of the citation's sentence, which closed all 9 skip cases. Two transient `judge_unavailable` errors in v1 (q-006, q-025) were both resolved by v2's retry-with-backoff; one new `judge_unavailable` appeared in v2 on q-013 (a correct draft), surviving all three retry attempts. The grounding rail in v2 therefore fires on 13/30 drafts — 3 of the 6 wrong (q-006, q-018, q-021) and 10 of the 24 correct.

Third, the v2 grounding rail's catches are substantive. Inspection of the three grounding flags on wrong drafts: (a) *q-006*: model attached the claim "ces prétentions doivent être suffisamment déterminées et ne pas être manifestement infondées" to ATF 119 II 337, but the cited source is about tenant/sub-tenant liability under Art. 101 CO; judge verdict `unrelated`, confidence high. c_{eli} judge agrees the draft is wrong (omits the 1-year deadline and consent rule). (b) *q-018*: model attached a claim about pension amount and contribution years to BGE 130 V 343, but that BGE is the LSE-Tabellenlöhne case, unrelated to pension calculation; judge verdict `no`. c_{eli} agrees the draft is wrong (uses the pre-2022 stepped scale). (c) *q-021*: model presented BGE 142 III 433 as about the maintenance damage of Art. 45 Abs. 3 OR, but the case is about *Schockschaden* (psychic impairment); judge verdict `unrelated`. c_{eli} agrees the draft is wrong (misses the Schockschaden component). All three v2 grounding flags on wrong drafts are clean *citation–claim mismatch* catches: the cited source does not support the claim attached to it, so the rail fires correctly. Whether each flag also identifies the draft's primary correctness failure varies: q-006's flag identifies a peripheral support claim, not the missing 1-year deadline that drove the c_{eli} verdict; q-018's flag is on a substantive doctrinal misattribution; q-021's flag identifies the central Versorgerschaden vs. Schockschaden confusion that drives the c_{eli} verdict. We do not claim the rail catches the primary error in every case — only that the firings are not artefacts.

The v2 trade-off is calibration, not coverage. The grounding rail also flags 10 of 24 correct-judged drafts, typically by registering a `partial` or `no` verdict on a side-claim that the c_{eli} judge accepted as part of the overall correct answer. Whether this is over-flagging by the grounding judge or excessive permissiveness by the c_{eli} judge is not something a same-family two-judge configuration can decide; v1.0 will substitute an independent judge to break the tie. The two wrong drafts the v2 rail still misses (q-028 and q-030) *do* reach the judge, which returns `partial` or `yes` on the extracted claim — in both cases the extracted claim is a narrow factual statement that is literally supported by the source even though the draft as a whole is wrong on the broader question. This is the *narrow-claim, broad-error* failure mode and joins calibration as the second open problem.

Cost / latency. Across the 30 drafts, the four deterministic rails together account for ~3 ms per draft (median 2.7 ms; 95 % under 6 ms), measured by wrapping the rail dispatch in `time.perf_counter_ns()` on the production deployment. The grounding rail takes 1.5–2.5 s wall-clock and ≈$0.005 per draft via the Anthropic API. Total experimental cost (generation + audit + judges): <$2.

6 Discussion

When the deterministic rails miss. The case-citation rail catches *fabricated* references, not *misused* ones. For systems where the underlying model has high citation-level fidelity (as Sonnet 4.6 has on Swiss law: ∼ 3 % fabrication rate across both runs), the deterministic rails contribute most of their value as a precision floor: they make sure the remaining ∼ 97 % of valid-looking citations are in fact valid. In v2, none of the four deterministic rails holds the 0 % FPR floor: C_1 (case-existence) still flags q-017's BGE 117 II 198 as fabricated, but the c_{eli} judge re-scored q-017 as correct in v2, so the same catch counts as 1 false positive (FPR 4.2 %). Adding statute, quote, and date raises combined FPR to 12.5 % by C_4. A production deployment can run all four cheaply, but the precision-floor argument depends on the ground-truth labelling — which our v1/v2 comparison shows is itself non-deterministic at this n.

When the judge rail miscalls. The grounding rail's solo FPR of 41.7 % in v2 (10/24 correct drafts flagged) plausibly overstates the production rate, but not for the reason we initially supposed: the rail already fetches the cited regeste / source text for verified citations, so the judge's access to source material does not change with retrieval. What changes with retrieval is the *generator*: a draft built on retrieved passages tends to anchor each claim with a pinpoint that maps cleanly to the source, leaving the judge less room to score `partial` or `unrelated`. This is a hypothesis, not a result of the present paper. The fundamental tradeoff remains in either condition: a stronger judge catches more reasoning errors but raises more disagreement-as-error verdicts; calibrating it — when to defer, when to override — is open work.

Generalisation. Each rail is corpus-agnostic in design. The case-existence rail needs an enumerable decision corpus; the statute rail needs a statute mirror; the quote rail needs the cited decisions' verbatim text. None of these are Swiss-specific. An English-law adaptation (e.g. England & Wales) would require: (i) regex patterns for English neutral citations and law-report references; (ii) a statute mirror for primary legislation (e.g. `legislation.gov.uk`-derived); (iii) decision full-text for the quote rail. Open or low-friction sources exist for much of this — BAILII for judgments, `legislation.gov.uk` for primary legislation — but ICLR / commercial law-report text sits behind licensing, and corpus-construction licensing is jurisdiction-specific in ways we do not attempt to inventory here. We expect the catch-rate / FPR profile to be model-dependent: a weaker generator with higher citation-level fabrication rate will see higher rail TPR.

Privacy and re-identification. The corpus we redistribute (969 k Swiss court decisions; packaging and added metadata under CC0; underlying texts are official published court decisions of the issuing courts) consists exclusively of decisions *as published by the issuing court* or via the official Swiss judicial publication portal `entscheidsuche.ch`. We perform no de-anonymisation, no re-identification, and no enrichment with non-public data. Nyffenegger, Stürmer & Niklaus (2024) measured LLM-based re-identification risk on anonymised Swiss court rulings and found it minimal for top frontier models on *court decisions* specifically (substantially higher on Wikipedia entities); we adopt their finding and discuss it explicitly in the dataset card. Our contribution to privacy is one-direction: we expose the already-public anonymised corpus through a more uniform interface, which does not create re-identification capacity beyond what was already feasible by direct scraping of the source portals. We do *not* mirror the in-court (pre-anonymisation) corpus; that work — including the operational *Anom2* system — belongs to the Niklaus, Mamié et al. (2023) line and the Federal Supreme Court itself.

Limitations.

- **End-to-end run variance.** Our v1 and v2 runs differ on 4 of 30 c_{eli} verdicts at TEMPERATURE = 0 (q-017 and q-026 went WRONG → CORRECT; q-006 and q-028 went CORRECT → WRONG). We do not separate generator variance from judge variance in this paper: every one of the 30 generated drafts also differed in text between the two runs, so the 4 flipped labels confound both sources. A held-fixed-drafts re-judging experiment would isolate the judge component; we defer it to v1.0 because the central finding — per-flip rate $\sim 13\,\%$ at $n = 30$ is large enough that any single run's ablation cannot be treated as the population value — holds either way. We report both runs side by side (Tables 1 and 2) so the structural findings can be inspected against both denominators. v1.0 will use an independent judge model and run multiple seeds.
- **Benchmark size.** v0.2 is a 30-question seed. With only 6 wrong drafts, the per-configuration WFR estimates are imprecise: 95 % Wilson confidence intervals on a 66.7 % point estimate ($n = 6$) are $[30.0\%, 90.3\%]$. We report exact ratios in Tables 1 and 2 so the reader can judge directly. v1.0 (150 questions, named expert co-authors from Swiss law faculties; see §4) is the planned next deliverable.
- **Single-curator benchmark.** v0.2's questions and reference answers were authored by the maintainer of `opencaselaw.ch`, a Swiss-trained author with practical legal-research experience but without the multi-annotator protocol that a published gold-standard benchmark requires. Selection bias is the obvious risk: a curator who built the underlying corpus may unintentionally choose questions whose evidence the corpus answers well. We mitigate by anchoring every evidence `decision_id` on the live corpus before publication and by publishing the evidence-resolution audit trail; we cannot eliminate the bias, and v1.0 addresses it directly with named expert co-authors and an independent annotation protocol.
- **Same model family, generator and judge.** Both roles are filled by Claude Sonnet 4.6 in our experiment. This concentrates failure modes (a wrong draft and a wrong-but-self-consistent judge will agree). v1.0 will adopt a SwiLTra-Judge-style independent calibrated judge (Niklaus et al., 2025) with human-expert validation, and report inter-judge agreement.
- **Ground-truth circularity.** We use the c_{eli} verdict (also produced by Sonnet 4.6) as the ground-truth label for which drafts are "wrong" in the per-rail ablation (Table 1). If Sonnet has a systematic bias on a given proposition, that bias affects both the draft and its c_{eli} judgment in the same direction, and a draft can be scored "correct" because the same misconception underlies both. We partially mitigate this by anchoring each question's reference answer on a corpus-verified BGE before generation, so the c_{eli} judge has an explicit external anchor to compare against; we do not claim to have eliminated the circularity. v1.0's independent judge addresses this directly.

- **Prior-only condition.** Our reported ablation runs the generator without retrieval, chosen as a stress-test of the audit pipeline (it maximises the density of recoverable errors), not as an end-to-end RAG evaluation. Butler & Butler (2026) report that retrieval is the primary determinant of legal-RAG performance; we do not contradict them, and an end-to-end retrieval-augmented evaluation against v1.0 is forthcoming. The audit pipeline is run on every production query at `mcp.opencaselaw.ch` (over a BM25 / RRF / Haiku-rerank retriever), but production-side audit catch-rates are not the subject of the present paper.

7 Conclusion

We propose a defense-in-depth closing audit pipeline for legal RAG, decomposing verification into five rails with bounded latency and cost, and report two runs of a 30-question stress test against Claude Sonnet 4.6 on Swiss federal law: a v1 run against the as-instrumented code, and a v2 run after closing the mechanical audit-pipeline gaps the v1 inspection identified. Three findings carry across both runs: citation existence is not the bottleneck (only 1 of 33 emitted citations is fabricated); the misrepresented- holding error class dominates the residual; and the c_{eli} judge has substantial run-to-run variance (∼13 % per-flip rate at $n = 30$). In v2 the grounding rail makes 3 clean semantic catches on wrong drafts (q-006, q-018, q-021) at FPR 41.7 %; calibration of the grounding judge, not coverage, is now the dominant open problem. Two wrong drafts (q-028, q-030) reach the judge but their extracted claims are narrow factual statements that the source literally supports — the *narrow-claim, broad-error* failure mode joins calibration as the second open problem.

The audit pipeline is live at `mcp.opencaselaw.ch`, deployed on the entire published Swiss federal and cantonal court corpus (969,000+ decisions, daily refresh). Code under MIT; benchmark packaging under CC0; underlying decision texts are official published court decisions of their issuing courts (with the collation, deduplication, and added metadata licensed CC0). We invite collaborators for v1.0.

Data and Code Availability

Reproducibility pin (2026-04-29). The numbers in §5 were produced from these exact artefacts. v1 (Table 2) was run against the audit code at commit `4553379`; v2 (Table 1) was run against the audit code at commit `12d32d3`, which adds citation-leading claim extraction and judge-call retry-on-error.

- **v2 audit pipeline source** (MIT), pinned to the v2 experiment commit:
  `https://github.com/jonashertner/caselaw-repo-1/blob/12d32d3/mcp_server.py`
- **v1 audit pipeline source** (MIT), pinned to the v1 experiment commit:
  `https://github.com/jonashertner/caselaw-repo-1/blob/4553379/mcp_server.py`
- Run harness (the script that generated 30 prior-only drafts and audited them):
  `https://github.com/jonashertner/caselaw-repo-1/blob/12d32d3/docs/paper/v2/experiments/01_run_ablation.py`
- Table-rendering script (computes `ablation_table.json/.md` from the per-trial results in post-processing):
  `https://github.com/jonashertner/caselaw-repo-1/blob/12d32d3/docs/paper/v2/experiments/02_compute_table.py`
- **v2 per-trial ablation data** (Table 1):
  `https://github.com/jonashertner/caselaw-repo-1/blob/12d32d3/docs/paper/v2/experiments_`

v2/ablation_results.json
SHA-256: 26deebb7...175d318b

- **v1 per-trial ablation data** (Table 2):
https://github.com/jonashertner/caselaw-repo-1/blob/4553379/docs/paper/v2/experiments/ablation_results.json
SHA-256: 79dcc442...039c3b08
- Derived ablation tables (the JSONs behind Tables 1 and 2):
https://github.com/jonashertner/caselaw-repo-1/blob/12d32d3/docs/paper/v2/experiments_v2/ablation_table.json
https://github.com/jonashertner/caselaw-repo-1/blob/12d32d3/docs/paper/v2/experiments/ablation_table.json
- Benchmark v0.2 (CC0): https://huggingface.co/datasets/voilaj/swiss-legal-rag-bench
SHA-256 of questions.jsonl: b299db37...f21ec449
- Corpus-verification audit trail (the evidence-resolution log produced when each question's BGE evidence was checked against the live corpus before publication):
https://github.com/jonashertner/caselaw-repo-1/blob/4553379/docs/paper/v2/03c_verification_results.md
- Underlying Swiss-caselaw corpus (packaging and added metadata under CC0; underlying texts are official published court decisions of the issuing courts): https://huggingface.co/datasets/voilaj/swiss-caselaw
Live MCP corpus size at v1 and v2 run time: 969,738 decisions. The HF dataset and the live MCP corpus are both updated daily as new decisions are scraped, so re-execution against the live endpoint will not be byte-exact: each resolved citation's decision_id is recorded in the per-trial JSON so a re-run can be compared against the stored values, and the published audit-pipeline code is byte-exact at the pinned commit, but the underlying corpus state drifts. We have not yet published a corpus snapshot keyed to the experiment date; a snapshot tarball would close the reproducibility gap and is on the v1.0 list.
- Live MCP endpoint (for re-execution against current corpus, with the drift caveat above):
https://mcp.opencaselaw.ch

Generator and judge: claude-sonnet-4-6 (Anthropic), temperature=0. Each trial in §5 can be re-executed by pinning to the commits + SHAs above and pointing the harness at the live MCP endpoint.

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